

EBU6304 Software Engineering Group Project

2025-2026

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First report

Requirements

This project is intended to develop a Teaching Assistant Recruitment System for BUPT International School. The current recruitment process mainly depends on manual forms and Excel files, which creates problems such as scattered information, unclear application status, and inefficient workload management. Therefore, our team used Agile methods to investigate stakeholder needs, define the project scope, and organise development work into iterations. For this first stage of the project, our goal is to complete the requirements investigation, prepare the product backlog, produce the prototype, and plan Sprint 1. At this stage, we mainly focus on the Minimum Viable Product (MVP), which means identifying and prioritising the essential core functions of the system before considering more advanced features. This helps the team focus on the most important user needs while keeping the first iteration realistic.

I. Requirements Investigation

To comprehensively understand stakeholder needs, the project team employed various requirements elicitation techniques.

1.1 Background Reading and Brainstorming

Before the formal investigation, the project team conducted background reading, including:

- Analyzing the existing TA recruitment process documents of the International School.

- Researching cases of similar TA management systems in other universities.

- Identifying pain points and limitations in the current Excel-based process.

Through brainstorming, the team preliminarily defined the system's core boundaries and main functional modules, laying the foundation for subsequent interviews and questionnaires.

1.2 Interview Method

To obtain more reliable and evidence-based requirements, the team carried out semi-structured interviews in Week 2 with 2 Module Organisers, 3 former Teaching Assistants, and 1 administrative staff member. Questions were tailored to stakeholder roles: MOs discussed vacancy publication and applicant review; former TAs shared experiences in tracking application progress; and administrative staff addressed workload balancing. The results confirmed the aforementioned pain points: applicant data is heavily fragmented, candidate comparison is time-consuming, and there is a lack of transparency for both students and administrators. These findings directly informed the backlog and led to the inclusion of core functions such as profile creation, CV upload, job posting, status tracking, applicant comparison, and workload summary.

1.3 Questionnaire Method

To quantitatively validate requirement priorities, an online questionnaire was distributed to 15 potential users (students intending to apply for TA positions).

Key Questionnaire Results:

87% of respondents were willing to use an online application system.

73% of respondents considered application status tracking a crucial function.

53% of respondents looked forward to a skill-matching function.

Summary of User Pain Points: No application receipt, unclear job requirements, difficulty finding suitable positions.

II. Project Glossary

2.1 Key Terms

Term	Definition
TA	Student applying for or holding a TA position.
MO	Personnel responsible for course management, including posting positions and screening applicants.
Admin	System administrator responsible for overall configuration and workload monitoring.
Position/Job	A specific TA work opportunity, including information such as course, responsibilities, and duration.
Application	A record of a TA's submission for a position.
Workload	The total hours of TA work undertaken by a specific TA within a given semester.

III. Requirements Prioritization

Based on the prioritization methods discussed in the course, we used the MoSCoW method to prioritize all user stories.

3.1 MoSCoW Prioritization

Must have: Core foundational functions essential for system launch (e.g., US01, US02, US03, US11).

Should have: Important but non-critical features (e.g., US04, US05, US08, US09).

Could have: Value-added functions implemented if time permits (e.g., US06, US07, US10).

Won't have: Features excluded from this iteration, such as payroll calculation and campus system integration.

3.2 Basis for Prioritization

1. Business Value: Must-have functions constitute the Minimum Viable Product (MVP) and can directly replace the existing Excel process.

2. User Feedback: Functions with high interest in the questionnaire were prioritized.

3. Technical Dependencies: Foundational functions (e.g., profile creation) were prioritized over dependent functions (e.g., job application).

4. Risk Consideration: High-risk functions were prioritized for early validation in initial iterations.

IV. Effort Estimation Method

Based on the estimation methods discussed in the course, we adopted Planning Poker for effort estimation, which is a team-consensus-based agile estimation technique.

The team adopted Planning Poker with Fibonacci story points because it was suitable for a six-member agile team and allowed effort estimation to be discussed collectively rather than

being dominated by a single opinion. This method was also appropriate for the early stage of the project, where some requirements were still evolving and the implementation details were not yet fully fixed. During estimation, each member independently selected a score after the user story was explained, and differences in scoring were then discussed until a shared understanding was reached. In practice, the team considered factors such as implementation complexity, data storage requirements, interface design effort, functional dependencies, and expected testing cost. For example, US01 Create TA Profile was estimated at 5 story points because it requires input forms, field validation, and persistent storage of profile data. US02 Upload CV was estimated at 3 story points because it mainly involves file path selection and linking the uploaded CV to an existing profile. US03 Post TA Job was estimated at 5 story points because it requires job information input, data persistence, and support for later application-related functions. In this way, the estimation results were not only based on general agile theory but also reflected the team's discussion of the actual development effort for Sprint 1.

V. Iterative Development Plan

Following the Scrum framework, we planned four 2-week iterations. Requirements from stakeholder interviews directly influenced backlog prioritization. For instance, due to urgent MO and TA needs, application status tracking was prioritized as a core function. Conversely, advanced features like AI skill matching were deferred to later sprints as they depend on the availability of structured data. Sprint 1 focuses strictly on the MVP: profile creation, CV upload, and job posting, ensuring the project remains feasible and aligned with the most urgent stakeholder needs.

5.1 Iteration Overview

Iteration	Timeline	Iteration Goal	Core Functions
Sprint 1	Weeks 3-4 (by Mar 22)	Build Core Foundational Functions	TA profile creation, CV upload, job posting
Sprint 2	Weeks 5-6 (by Apr 12)	Complete the Application Process	Job application, screening, withdrawal, job management
Sprint 3	Weeks 7-9 (by May 3)	Add Management Functions	Workload summary, report export, status notifications
Sprint 4	Weeks 10-11 (by May 17)	Smart Features & System Optimization	AI skill matching, AI suggestions, system logs, testing & documentation

5.2 Sprint 1 Detailed Plan (Deliverable for First Assessment)

ID	User Story	Acceptance Criteria	Story Points
US01	Create TA Profile	Fields: name, student ID, email, skills; Save to .txt or .json.	5
US02	Upload CV	Specify file path; path stored in association with the profile.	3
US03	Post TA Job	Fields: course name, requirements, deadline; Data persistence.	5
Total			13

TSprint 1 is the primary focus of this submission, representing the project's current stage and MVP feasibility. Following the staged requirements of the assessment, subsequent sprints are outlined briefly as future development plans.

5.3 Iteration Management Process

Daily Stand-up Meeting: 15-minute syncs to track progress and remove blockers.

Sprint Review: Stakeholder demonstrations to gather functional feedback.

Sprint Retrospective: Team reflection to optimize process for the next iteration.

VI. Prototype Design

A medium-fidelity prototype was developed to validate layout, user flows, and MVP feasibility. It demonstrates core structure and conceptual features rather than a fully implemented system.

6.1 Prototype Design Principles

Demonstrate the main user flows for TA applicants and Module Organizers. Reflect the overall interface layout, page navigation, and role-based entry. Show key interactions such as profile viewing, job browsing, application status tracking, and applicant review.

6.2 Main Interfaces

Login Interface: role selection for TA, MO, and Admin users.

TA Dashboard: Profile management, job browsing, and status tracking.

MO Management Interface: Job posting, CV review, and application decision-making.

Note: AI-related features are conceptualized in the prototype but scheduled for later sprints.

VII. Supporting Materials

The following materials provide evidence of stakeholder feedback and requirement validation, ensuring the backlog and iteration plans are grounded in actual investigation results.

7.1 Interview Findings Summary

Evidence Type	Description	Impact on Design
Interview note – Module Organizers	Module Organizers reported that the current recruitment process relies heavily on forms and Excel files, making it difficult to compare applicants efficiently.	Added applicant comparison and job management related requirements.
Interview note – Former TAs	Former TAs stated that they often could not clearly track their application progress after submission.	Prioritized application status tracking in the backlog and prototype.
Interview note – Administrative Staff	Administrative staff highlighted the need to monitor TA workload across different courses.	Added workload summary and cross-course overview as later management functions.

7.2 Questionnaire Result Summary

Questionnaire Result	Evidence	Impact on Design
Willingness to use online system	87% of respondents were willing to use an online TA application system.	Confirmed the feasibility and user acceptance of the proposed system.
Importance of status tracking	73% of respondents considered application status tracking very important.	Increased the priority of status tracking as a core user-facing feature.
Interest in skill matching	53% of respondents expressed strong interest in a skill-matching function.	Retained AI skill matching in the prototype but scheduled it for later iterations rather than Sprint 1.
Preferred application method	80% preferred online forms.	Supported the decision to design a structured digital application flow.

7.3 Early Feedback Summary

Evidence Type	Description	Impact on Design
Prototype feedback – user role entry	Test users suggested that different stakeholders should see different entry points after login.	Added role selection in the login interface for TA, MO, and Admin users.
Prototype feedback – application visibility	Users wanted to check submitted applications in one place rather than through repeated enquiries.	Included the “My Applications” page in the TA dashboard.
Prototype feedback – management overview	Stakeholders preferred a simple dashboard view for monitoring workload and job records.	Added workload dashboard and export-related functions to the admin interface.

7.4 Planning Notes

Evidence Type	Description	Impact on Planning
MoSCoW discussion record	The team agreed that profile creation, CV upload, and job posting were essential for launching the MVP.	These stories were assigned to Sprint 1 as must-have functions.
Dependency analysis note	Advanced features such as AI matching depend on structured applicant and job data being available first.	AI-related features were moved to later iterations.
Scope control note	Payroll calculation and campus system integration were considered outside the current MVP scope.	These functions were excluded from the first assessment backlog.

7.5 Client Feedback

The client mentioned that the current Excel-based process is difficult to manage. Applicant information needs to be compiled manually, and repeated enquiries about application status create extra workload.

This feedback further supports the need for centralized applicant records, application status tracking, and more efficient management functions in the proposed system.